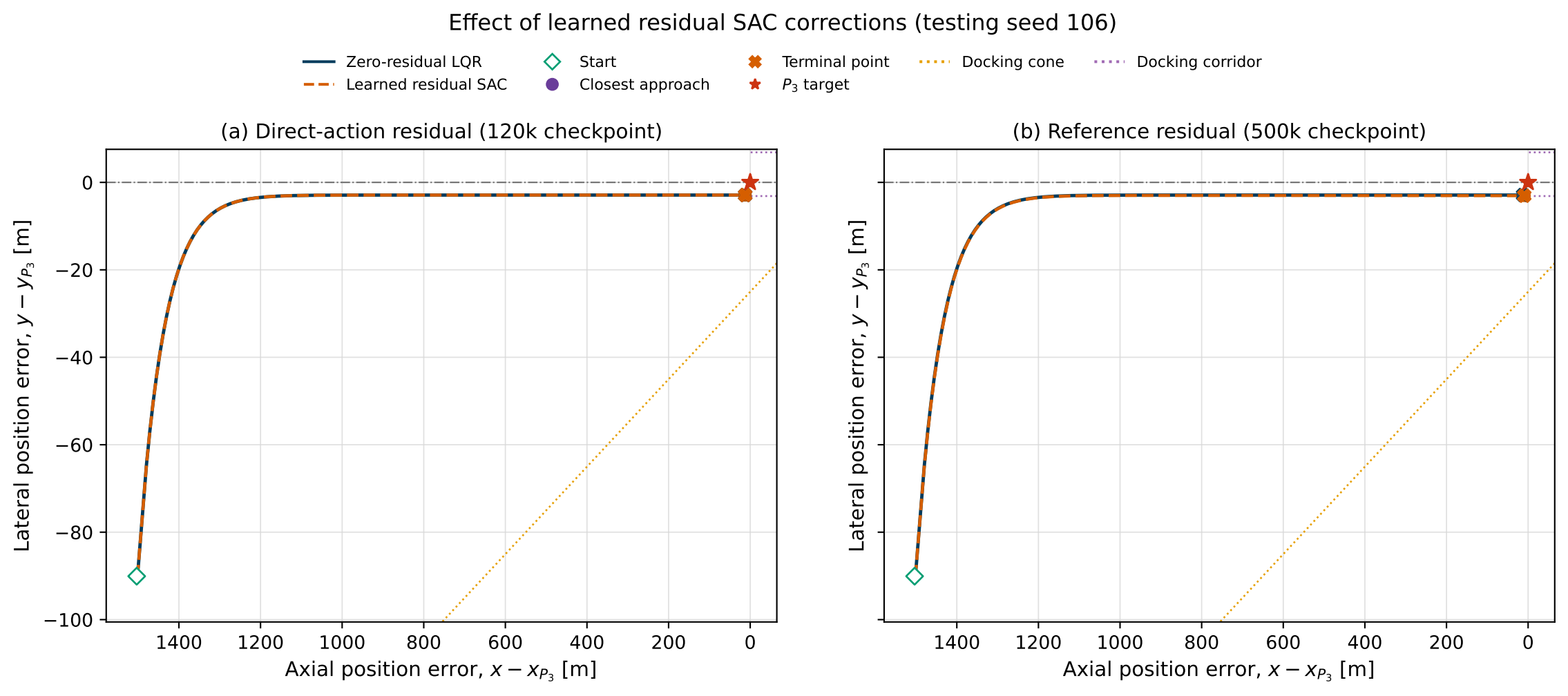
Effect of learned residual SAC corrections (testing seed 106)
Zero-residual LQR
Learned residual SAC
Start
Closest approach
Terminal point
P_3 target
Docking cone
Docking corridor
(a) Direct-action residual (120k checkpoint)
(b) Reference residual (500k checkpoint)
Lateral position error, $y - y_{P_3}$ [m]
Axial position error, $x - x_{P_3}$ [m]
0
−20
−40
−60
−80
−100
1400
1200
1000
800
600
400
200
0